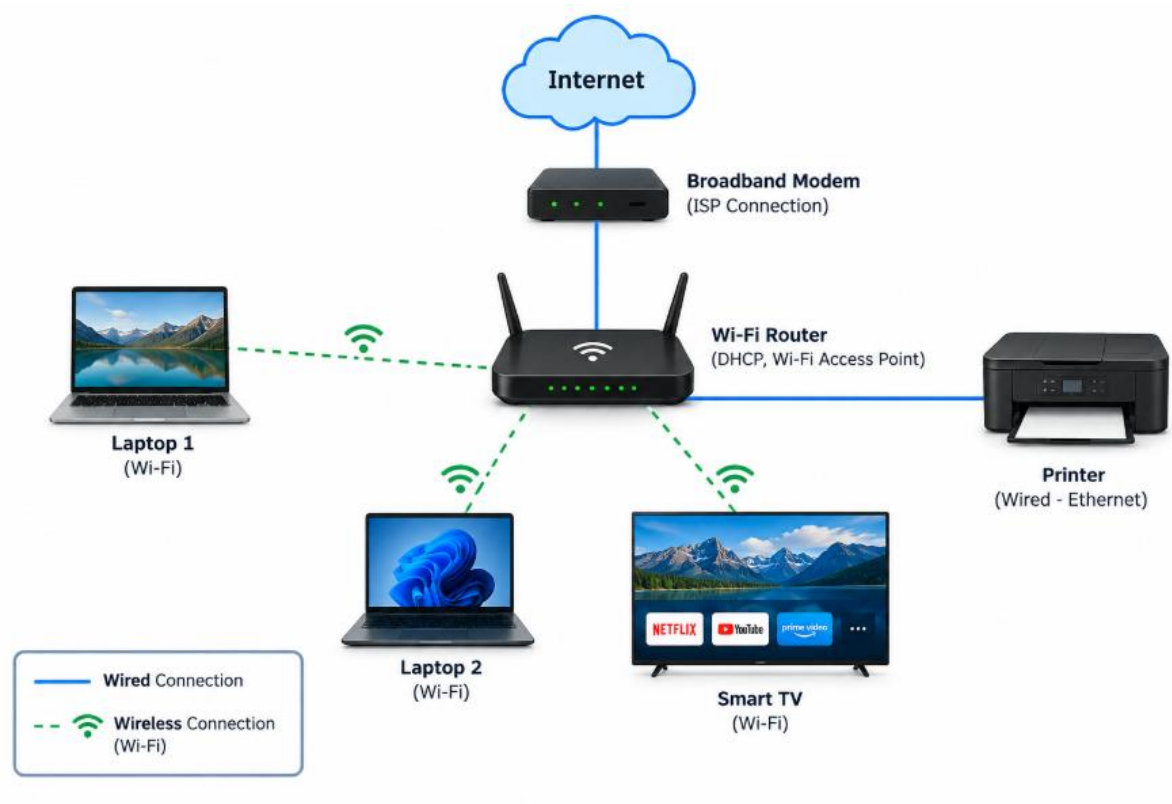
Task 1:

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

Answer:



SOHO Network Diagram Explanation:

This diagram represents a small office/home office (SOHO) network. The broadband connection provides Internet access through the Wi-Fi router. The router connects two laptops and a Smart TV using wireless (Wi-Fi) connections. The printer is connected to the router using a wired Ethernet connection. The Wi-Fi router manages communication between all connected devices and provides Internet access to them.

Task 2:

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Answer :

Hardware Components Required:

1. **Broadband Internet Connection**
    - Any local ISP such as JioFiber, Airtel Xstream Fiber, or BSNL Fiber.
    - The ISP normally provides the ONT/modem required for the broadband connection.
2. **Wi-Fi Router**
    - Suggested model: TP-Link Archer C6 AC1200
    - Dual-band Wi-Fi (2.4 GHz and 5 GHz)
    - 1 WAN port and 4 Gigabit LAN ports.
    - Suitable for connecting laptops, Smart TV and printer.
3. **Laptop 1 and Laptop 2**
    - Any Wi-Fi-enabled laptops can be used.
    - Example brands: HP, Dell, Lenovo, ASUS.
    - No Ethernet cable is required because they connect through Wi-Fi.
4. **Smart TV**
    - Any Wi-Fi-enabled Smart TV.
    - Example brands: Samsung, LG, Sony, Xiaomi.
    - Connects wirelessly to the Wi-Fi router.
5. **Network Printer**
    - A network/Wi-Fi printer can be used.

- Example: Canon PIXMA TS307/TS307A or a similar Wi-Fi printer.
- If using a wired printer as shown in the diagram, the printer should have an Ethernet port.

6. **Ethernet/LAN Cable**
    - Cat6 RJ-45 Ethernet cable
    - Needed for:
        - Broadband modem/ONT → Wi-Fi router
        - Wi-Fi router → wired printer
    - Keep 2 Cat6 cables of suitable length.

Task 3:

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Answer:

Step-by-Step Wi-Fi Router Configuration

1. Connect the Router
    - Connect the broadband modem/ONT to the router's WAN/Internet port using an Ethernet cable.
    - Turn on the modem and Wi-Fi router.
    - Connect a laptop to the router using Wi-Fi or an Ethernet cable.
2. Open the Router Login Page
    - Open a web browser.
    - Enter the router's default IP address, such as 192.168.0.1 or 192.168.1.1.
    - Enter the default administrator username and password given on the router label/manual.
3. Change the Admin Password
    - Go to Administration / System Tools → Password.
    - Change the default admin password to a strong and unique password.
    - Save the changes.
    - This prevents unauthorized users from accessing the router's settings.
4. Set a Unique SSID
    - Open Wireless / Wi-Fi Settings.
    - Change the default Wi-Fi name (SSID) to a unique name, for example SOHO_Home_Network.
    - Avoid using personal information such as your name or phone number.
    - Save the settings.
5. Enable WPA2 Security
    - Go to Wireless Security settings.

- Select WPA2-PSK (AES) as the security/encryption method.
- Enter a strong Wi-Fi password containing a combination of uppercase letters, lowercase letters, numbers, and special characters.
- Save/apply the settings.

6. Configure Internet Settings
    - Go to the Internet/WAN settings.
    - Select the connection type provided by your ISP, such as Dynamic IP (DHCP), PPPoE, or Static IP.
    - Enter ISP-provided details if required.
    - Save the configuration.
7. Reconnect Devices
    - After applying the settings, the router may restart.
    - On both laptops, Smart TV, and other Wi-Fi devices, search for the new SSID.
    - Enter the new Wi-Fi password to connect them.
8. Test the Network
    - Open a website on the laptops or Smart TV.
    - Verify that all devices can access the Internet.
    - Check that the printer is also connected and accessible.

Result

The SOHO Wi-Fi router is now configured with a unique SSID, WPA2 security, and a new administrator password, providing a more secure wireless network.

Task 4:

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Answer:

Device	Connection	MAC Address	IP Address	Assignment
Wi-Fi Router	Wired	00:1A:2B:3C:4D:01	192.168.1.1	Static
Laptop 1	Wireless	00:1A:2B:3C:4D:11	192.168.1.100	DHCP
Laptop 2	Wireless	00:1A:2B:3C:4D:12	192.168.1.101	DHCP
Smart TV	Wireless	00:1A:2B:3C:4D:20	192.168.1.20	Static
Printer	Wired	00:1A:2B:3C:4D:30	192.168.1.10	Static

Network Details

- Network: 192.168.1.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.1
- DHCP Range: 192.168.1.100 – 192.168.1.200
- Printer IP: 192.168.1.10
- Smart TV IP**:** 192.168.1.20

Note: The printer and Smart TV use static IP addresses so their addresses remain fixed. The two laptops obtain their IP addresses automatically from the router using DHCP.

Task 5:

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Answer:

If one laptop cannot connect to the Internet while other devices are working, I would troubleshoot it step-by-step as follows.

1. Wi-Fi Connection Problem

Possible cause: The laptop may be disconnected from Wi-Fi or connected to the wrong SSID.

Steps to diagnose and fix:

1. Check whether Wi-Fi is turned ON on the laptop.
2. Check if the laptop is connected to the correct SSID, such as SOHO_Home_Network.
3. Disconnect and reconnect to the Wi-Fi network.
4. Re-enter the Wi-Fi password if required.
5. If it still does not connect, restart the laptop and router.
6. Check whether the laptop can connect to another Wi-Fi network to determine whether the problem is with the laptop or router.

2. Incorrect IP Address or DHCP Problem

Possible cause: The laptop may not have received a valid IP address from the router.

Steps to diagnose and fix:

1. Open Command Prompt on the laptop.
2. Run: ipconfig
3. Check whether the laptop has an IP address such as 192.168.1.100.
4. Check the Default Gateway. It should be 192.168.1.1.

5. If the IP address is missing or incorrect, run:

   ipconfig /release

   ipconfig /renew

6. Run: ping 192.168.1.1
7. If the router does not respond, check the Wi-Fi connection and DHCP settings on the router.

3. DNS Problem

Possible cause: The laptop may have Internet connectivity, but DNS may not be resolving website names.

Steps to diagnose and fix:

1. First test the Internet using: ping 8.8.8.8
2. If this works but: ping google.com

   fails, the problem may be DNS.

3. Check the DNS settings using:

   ipconfig /all

4. Restart the network adapter or renew the DHCP lease.
5. If necessary, configure a reliable DNS server such as 8.8.8.8 or 1.1.1.1.
6. Test the connection again.

4. Router or Internet Connection Problem

Possible cause: The router or broadband connection may have lost Internet access.

Steps to diagnose and fix:

1. Check whether other devices can access the Internet.
2. Check the router's Internet/WAN status.
3. Restart the broadband modem/ONT and router.

4. Check that the Ethernet cable between the modem/ONT and router is properly connected.
5. If all devices lose Internet access, contact the ISP if the problem continues.